

Physics 101  
**Lab 13: Hooke's Law**  
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## Purpose

The purpose of this thirteenth lab is to obtain practical skills in performing an experiment to attest the Hooke's law through the use of two springs.

## Members

This lab group consisted of four members who worked together as a team to make the procedure accurate and fair for everyone.

1. Myat Thit Ko Ko (author)
2. Sarah Ruelas
3. Inigo Mikael Dela Vega
4. Hector Torres

## Procedure

The lab consisted of the following steps and procedures.

### Part I: Spring Constants

- Stacking weights, measuring height difference, producing a graph with slope K value
- Timing 10 oscillations, calculating K from slope obtained with period squared against x

### Part II: Springs in Series

- Stacking weights, measuring height difference, producing a graph with slope K value

### Part III: Springs in Parallel

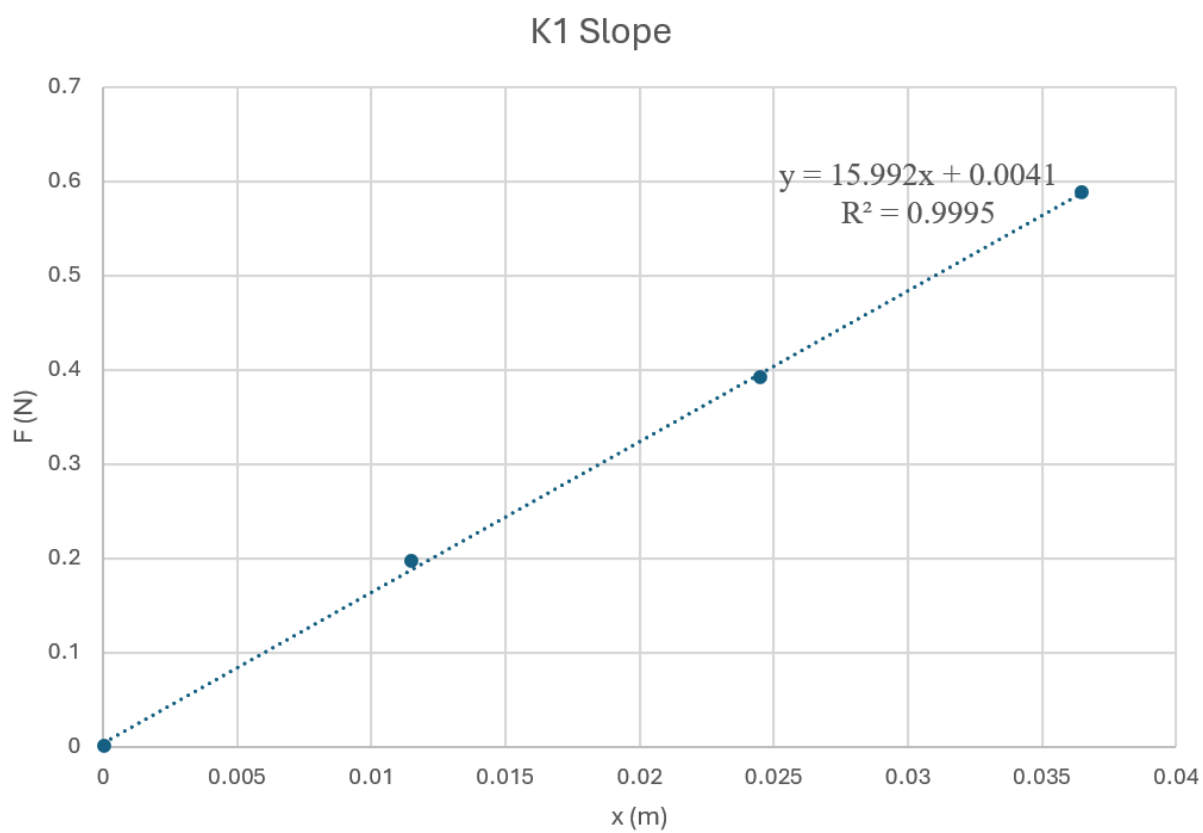
- Stacking weights, measuring height difference, producing a graph with slope K value

## Data & Analysis

### Part I: Spring Constants

#### K1

| m (kg) | x (m)  | F (N) |
|--------|--------|-------|
| 0.020  | 0.0115 | 0.196 |
| 0.040  | 0.0245 | 0.392 |
| 0.060  | 0.0365 | 0.588 |

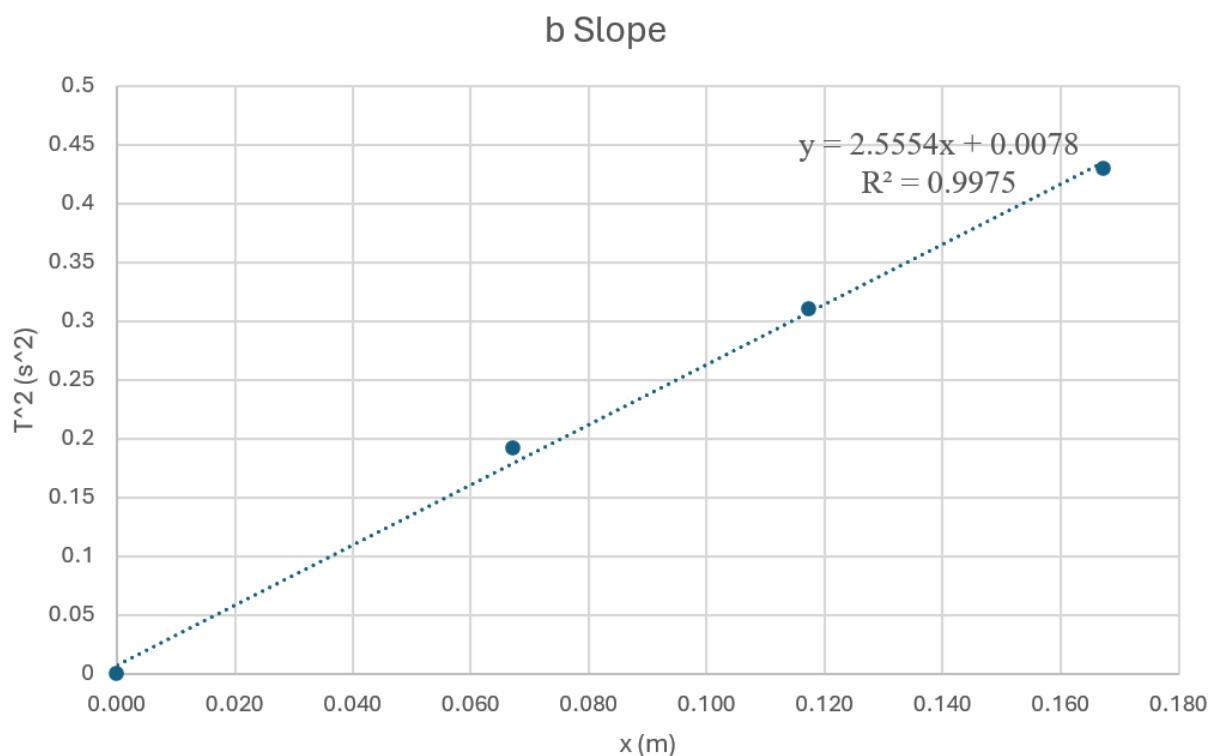


$$K_{1\text{exp}} = 16.0 \text{ Nm}$$

## K1 Oscillation

Mass of hanger = 0.0173 kg

| stacks (kg) | m (kg) | t for 10 oscil | T(s)   | T <sup>2</sup> (s <sup>2</sup> ) |
|-------------|--------|----------------|--------|----------------------------------|
| 0.05        | 0.067  | 4.38           | 0.438  | 0.191844                         |
| 0.1         | 0.117  | 5.565          | 0.5565 | 0.309692                         |
| 0.15        | 0.167  | 6.55           | 0.655  | 0.429025                         |

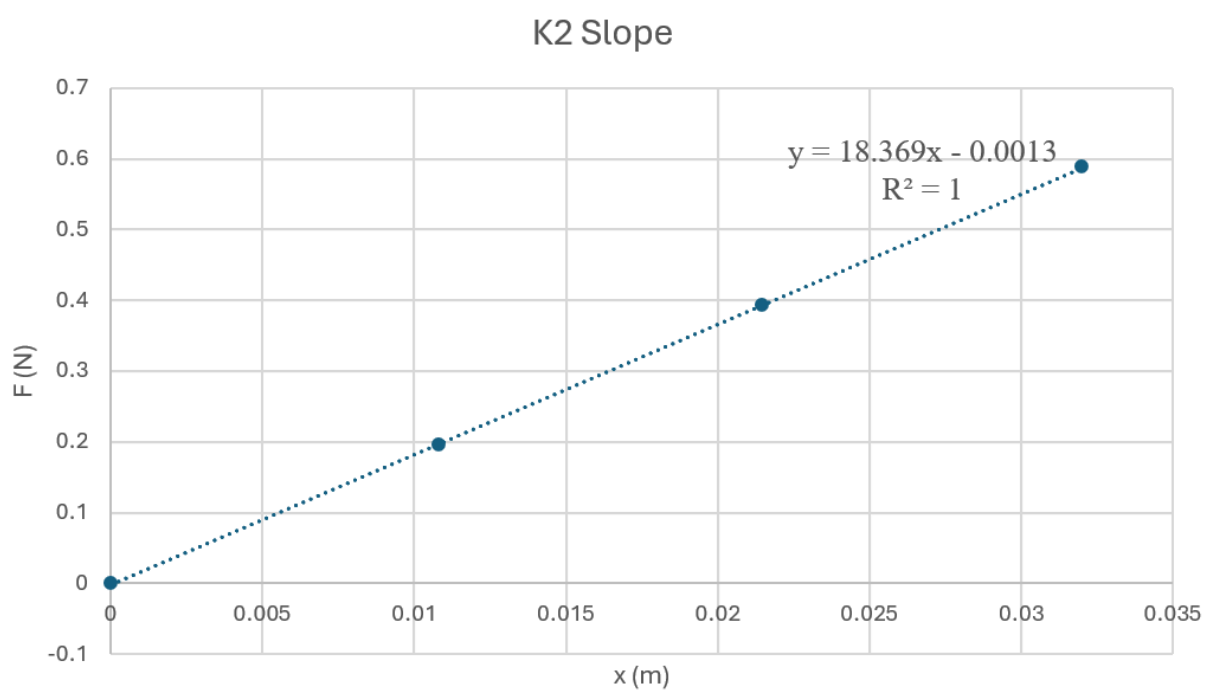


$$K_{1\text{theo}} = \frac{(4\pi^2)}{b} = \frac{(4\pi^2)}{(2.56)} = 15.4 \text{ Nm}$$

$$\text{error \%} = \frac{|K_{1\text{theo}} - K_{1\text{exp}}|}{\frac{K_{1\text{theo}} + K_{1\text{exp}}}{2}} \times 100\% = \frac{|15.4 \text{ Nm} - 16.0 \text{ Nm}|}{\frac{15.4 \text{ Nm} + 16.0 \text{ Nm}}{2}} \times 100\% = 3.82 \%$$

**K2**

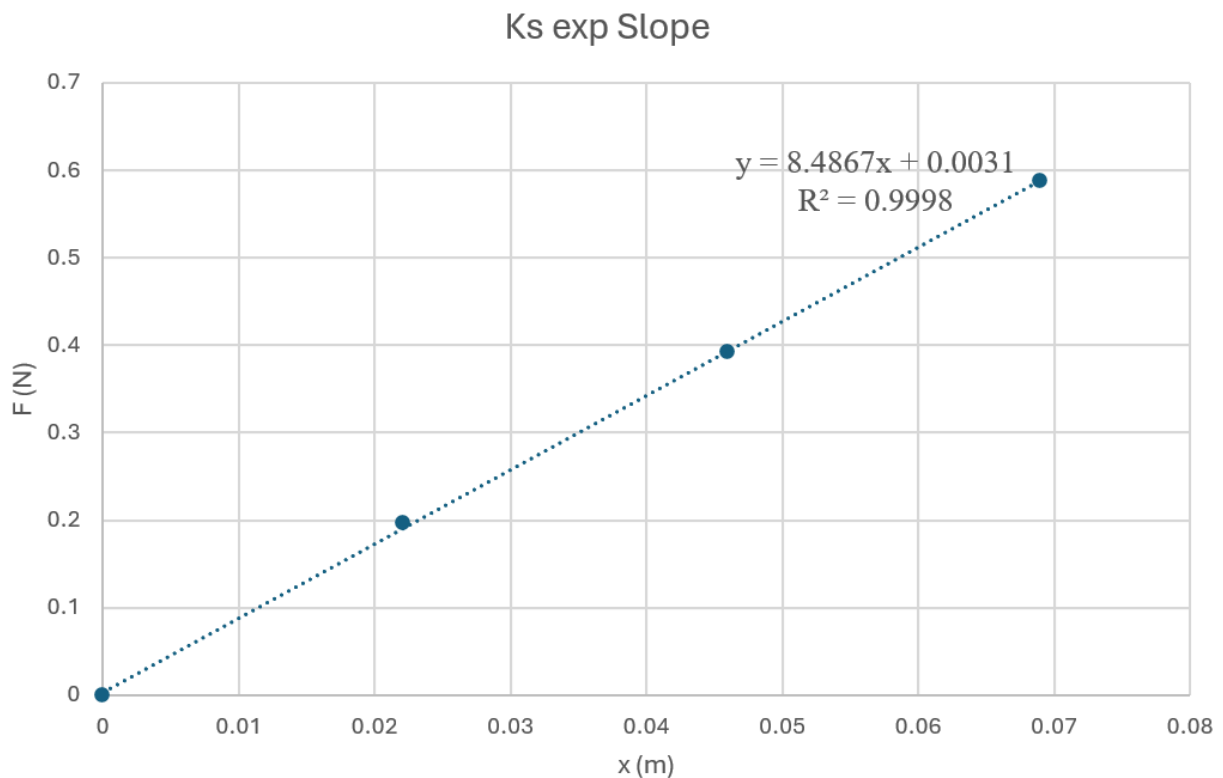
| m (kg) | x (m)  | F (N) |
|--------|--------|-------|
| 0.020  | 0.0108 | 0.196 |
| 0.040  | 0.0215 | 0.392 |
| 0.060  | 0.032  | 0.588 |



$$K_2 = 18.4 \text{ Nm}$$

## Part II: Springs in Series

| m (kg) | x (m)  | F (N) |
|--------|--------|-------|
| 0.020  | 0.0221 | 0.196 |
| 0.040  | 0.046  | 0.392 |
| 0.060  | 0.069  | 0.588 |



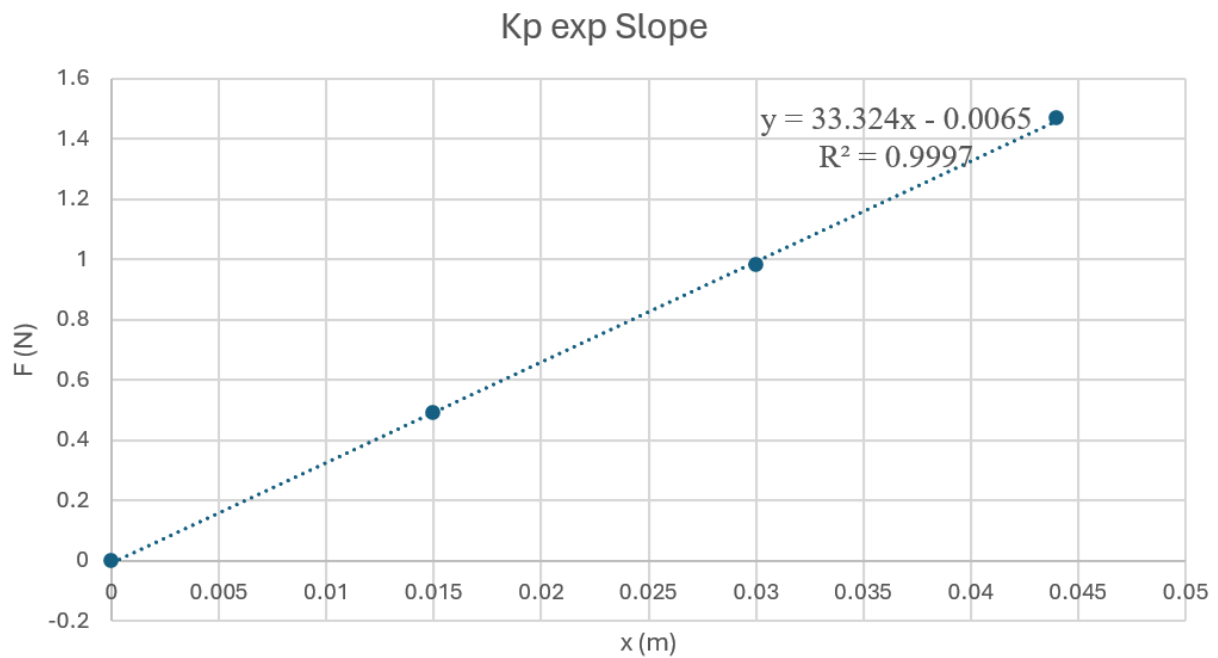
$$K_{s \text{ exp}} = 8.49 \text{ Nm}$$

$$K_{s \text{ theo}} = \left( \frac{1}{K_1} + \frac{1}{K_2} \right)^{-1} = \left( \frac{1}{16} + \frac{1}{18.4} \right)^{-1} = 8.56 \text{ Nm}$$

$$\text{error \%} = \frac{|K_s^{\text{theo}} - K_s^{\text{exp}}|}{\frac{K_s^{\text{theo}} + K_s^{\text{exp}}}{2}} \times 100\% = \frac{|8.56 \text{ Nm} - 8.49 \text{ Nm}|}{\frac{8.56 \text{ Nm} + 8.49 \text{ Nm}}{2}} \times 100\% = 0.821\%$$

### Part III: Springs in Parallel

| m (kg) | x (m) | F (N) |
|--------|-------|-------|
| 0.050  | 0.015 | 0.49  |
| 0.100  | 0.03  | 0.98  |
| 0.150  | 0.044 | 1.47  |



$$K_{p \text{ exp}} = 33.3 \text{ Nm}$$

$$K_{p \text{ theo}} = K_1 + K_2 = 16 \text{ Nm} + 18.4 \text{ Nm} = 34.4 \text{ Nm}$$

$$\text{error \%} = \frac{|K_p^{\text{theo}} - K_p^{\text{exp}}|}{\frac{K_p^{\text{theo}} + K_p^{\text{exp}}}{2}} \times 100\% = \frac{|34.4 \text{ Nm} - 33.3 \text{ Nm}|}{\frac{34.4 \text{ Nm} + 33.3 \text{ Nm}}{2}} \times 100\% = 3.25\%$$



## Analysis

All error percentages were below 5%. Springs in Parallel have a much higher error percentage compared to springs in series. This is likely due to the difficulty of setting up perfectly parallel springs whereas connecting springs in series was relatively easy.

## Conclusion

Since error percentages were found to be less than 10%, this lab is deemed satisfactory.

## Summary

Experiment Title: Hooke's Law

Student's name: Myat Thit Ko Ko

Date: 05/14/2026

Purpose: To attest Hooke's Law

Procedure: Stacking weights, timing oscillations, measuring spring constants

Data:  $5 \times 3 = 15$  sets total

Analysis: Error percentages were 3.82%, 0.821%, and 3.25% respectively.

Conclusion: Satisfactory